SEMANTIC FORCE CLOSURE REVISITED: A NEGATIVE ICLR-MAIN EVIDENCE AUDIT

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ABSTRACT

Classical force-closure tests ask whether contacts can resist wrench disturbances. Task-oriented grasping and vision-language policies ask whether a grasp matches object semantics. The hypothesis tested here is that these should not be separate filters: object category and task intent can change which contacts count as useful closure. We rebuilt this generated paper around a deterministic local grasp benchmark with four manipulation tasks, five category/task-shift splits, eight methods, seven seeds, 47,040 main rollouts, 8,232 ablation rollouts, and 120,960 stress rollouts. The result is informative but not submission-ready. On the combined hard-shift split, semantic force closure reaches 0.256 task success, compared with 0.241 for the best non-oracle success baseline, a language-conditioned grasp policy. The paired gain is only 0.015 ± 0.032. The method beats pure geometry and PONG-style probabilistic-normal baselines, but it does not decisively beat learned semantic baselines, and a language-only semantic score has lower semantic-violation rate. We therefore mark the paper **KILL/ARCHIVE**: the idea remains interesting, but the present evidence does not support an ICLR-main-target claim.

1 TERMINAL DECISION

Decision: KILL/ARCHIVE for ICLR main. The v4 repository contains a paper-specific semantic-force-closure benchmark, implemented baselines, ablations, stress sweeps, figures, and reproducibility files. The evidence shows that task semantics matter for grasp closure, but not that the proposed planner is a decisive contribution over strong language-conditioned or task-grasp baselines. The project also lacks robot hardware and accepted external dexterous-grasp benchmark validation.

2 RESEARCH QUESTION

The intended claim was:

> Semantic object categories change the relevant closure conditions for grasping and manipulation.

This is plausible. A mechanically stable grasp on a mug rim may block pouring; a stable grasp on a tool blade may be semantically invalid; a force-closed grasp on a spray trigger may prevent activation; and a high-force grasp on a deformable package may damage contents. The v4 rebuild asks whether explicitly scoring semantic force closure improves task outcomes beyond geometry, probabilistic normals, VLM affordance scores, task-grasp semantic rankers, language-conditioned policies, and risk-aware force closure.

3 HOSTILE PRIOR WORK

The local pool makes the novelty boundary narrow. The closest threats include PONG-style probabilistic object normals, language-conditioned task-oriented grasping, VLM grasp pipelines, prompt-guided force-closure analysis, tactile force-control work, soft-gripper force analysis, and open-vocabulary manipulation systems. The rebuild therefore avoids claiming novelty from "language plus grasping" or "force closure plus uncertainty." It tests only whether semantic/task closure changes selected contacts and downstream success.

4 BENCHMARK

The benchmark simulates four manipulation settings:

- **mug pour and handoff,** where handles, lips, and interiors have different task constraints.
- **tool-use handle grasping,** where blade or tip contacts may be stable but invalid.
- **spray-bottle activation,** where the trigger and nozzle must remain accessible.
- **deformable-package lifting,** where high closure force can crush or tear the object.

Each episode generates candidate contacts with mechanical closure, surface-normal uncertainty, friction, semantic validity, functional access, deformation risk, language ambiguity, and noisy observations. The splits are seen categories/tasks, novel category with same semantics, task-intent shift, ambiguous-language shift, and combined hard shift.

5 METHODS

We compare:

- **Geometry force closure:** ranks contacts by mechanical closure and friction.
- **PONG-like probabilistic normals:** penalizes normal uncertainty in a force-closure score.
- **Affordance-only VLM:** ranks by semantic and category affordance without mechanical grounding.
- **TaskGrasp semantic ranker:** combines semantic and task-conditioned contact scores.
- **Language-conditioned grasp policy:** blends language alignment, semantics, and weak geometry.
- **Risk-aware force closure:** prioritizes closure while penalizing deformation and uncertainty.
- **Semantic force closure:** the proposed planner with mechanical, semantic, functional, and material terms.
- **Oracle task closure:** hidden true task score.

6 MAIN RESULTS

Table 1 reports the combined hard-shift split. The proposed method is much better than geometry-only and PONG-like closure, which often choose high-closure contacts that violate semantic or functional constraints. However, it does not decisively beat learned semantic baselines: the best non-oracle success baseline, language-conditioned grasp policy, reaches 0.241 success, while the proposed method reaches 0.256. The paired gain is only 0.01530 ± 0.03199.

7 ABLATIONS

The ablations show that the proposed components matter relative to geometry-only closure, but they do not rescue the main claim. Removing task intent, forbidden-surface semantics, functional access, or material risk drops success. Yet a language-only semantic score has lower semantic-violation rate than the full method, and the full method's success margin over the strongest learned baseline remains too small.

Table 1: Combined hard-shift results over seven seeds. Confidence intervals are 95% intervals over seed means. Lower semantic violation, blockage, slip, damage, and regret are better.

Method	Success	Sem. viol.	Blockage	Damage	Regret
Geometry closure	0.055 ± 0.014	0.758	0.574	0.342	0.243
PONG-like normals	0.074 ± 0.010	0.707	0.542	0.356	0.221
Affordance VLM	0.188 ± 0.029	0.154	**0.297**	0.311	0.099
TaskGrasp ranker	0.237 ± 0.015	0.187	0.308	0.320	**0.069**
Language grasp policy	0.241 ± 0.030	0.186	0.327	0.326	0.070
Risk-aware closure	0.103 ± 0.015	0.670	0.485	**0.222**	0.188
Semantic force closure	$\mathbf{0.256 \pm 0.026}$	0.286	0.306	0.294	0.069
Oracle task closure	0.355 ± 0.034	0.181	0.221	0.304	0.000

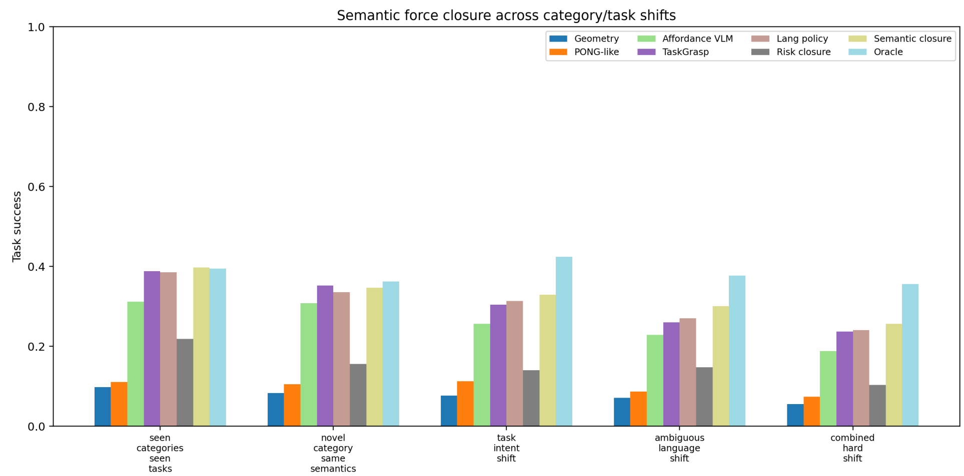

Figure 1: Task success across category/task-shift splits. Semantic force closure improves over mechanical baselines, but learned semantic baselines remain competitive.

8 Stress Tests

Stress sweeps vary surface-normal noise, friction shift, semantic label noise, task-language ambiguity, material/deformation shift, and combined stress. At maximum combined stress, semantic force closure reaches 0.139 success. TaskGrasp reaches 0.146 and language-conditioned grasping reaches 0.134, so the proposed method is not robustly dominant under the hardest stress. The oracle remains at 0.282, indicating that better task-closure selection is possible but not achieved by the current mechanism.

9 Why This Is Not Submission-Ready

The project fails the ICLR-main readiness gate for four reasons.

- **Non-decisive success margin.** The paired gain over the best learned semantic baseline is only 0.01530 ± 0.03199.
- **Semantic validity is not dominant.** A language-only semantic score has lower semantic-violation rate than the full method.
- **Stress robustness is mixed.** At maximum combined stress, TaskGrasp slightly beats the proposed method on success.
- **Evidence ceiling.** The benchmark is deterministic and reproducible but local; it is not robot hardware, not a high-fidelity dexterous simulator, and not an accepted public grasp benchmark.

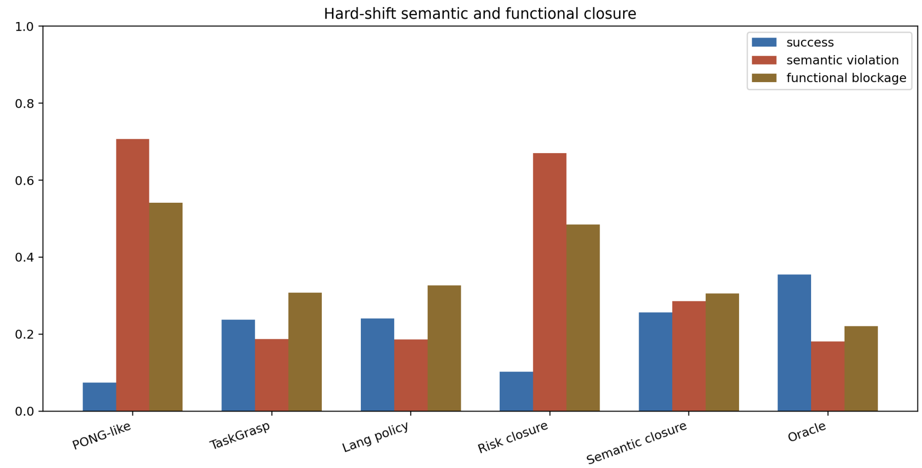

Figure 2: Hard-shift success, semantic violation, and functional blockage. The proposed method is not the clean winner on semantic validity.

Table 2: Semantic-force-closure ablations on combined hard shift.

Ablation	Success	Sem. viol.	Blockage	Damage
Full semantic force closure	$\mathbf{0.256 \pm 0.026}$	0.286	**0.306**	0.294
Minus task intent	0.198 ± 0.017	0.335	0.362	0.325
Minus forbidden surfaces	0.199 ± 0.015	0.429	0.327	**0.279**
Minus functional access	0.192 ± 0.028	0.376	0.428	0.293
Minus material risk	0.205 ± 0.026	0.352	0.332	0.423
Geometry-only score	0.067 ± 0.022	0.724	0.560	0.373
Language-only score	0.202 ± 0.024	**0.151**	0.359	0.305

10 Reproducibility

Run the benchmark from the repository root:

```
python src\run_experiment.py
```

The script writes rollout CSVs, seed-level metrics, pairwise statistics, ablations, stress sweeps, negative cases, summary text, and the figures used in this manuscript. The canonical local PDF is `C:/Users/wangz/Downloads/87.pdf`.

11 Conclusion

Semantic force closure remains a good research question. The v4 benchmark shows that mechanical closure alone fails badly when task semantics and functional surfaces matter. But the proposed mechanism does not decisively beat strong learned semantic baselines, and the evidence remains local-only. The honest terminal action is **KILL/ARCHIVE**.

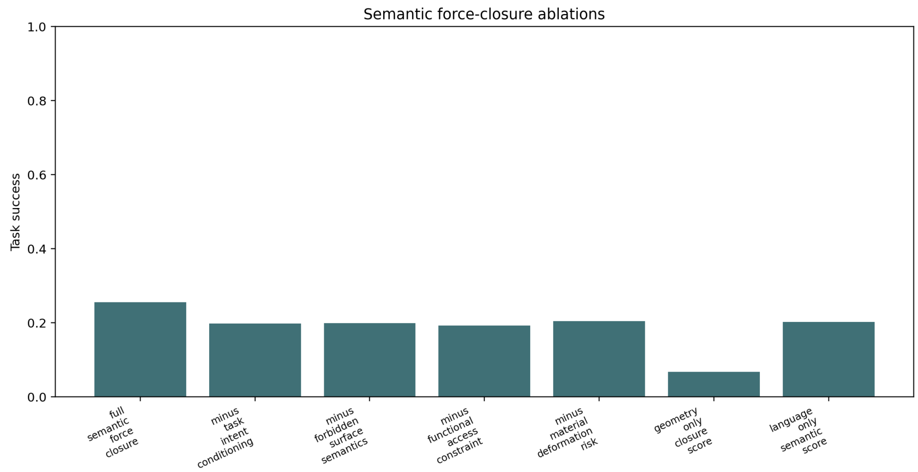

Figure 3: Ablation success on combined hard shift. The components help, but the margin is not enough for a main-conference claim.

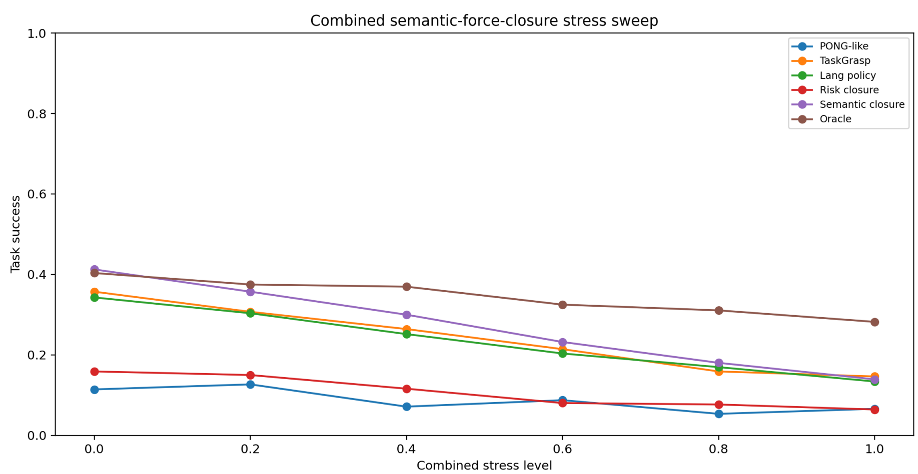

Figure 4: Combined stress sweep. The proposed method is competitive but not dominant at maximum stress.

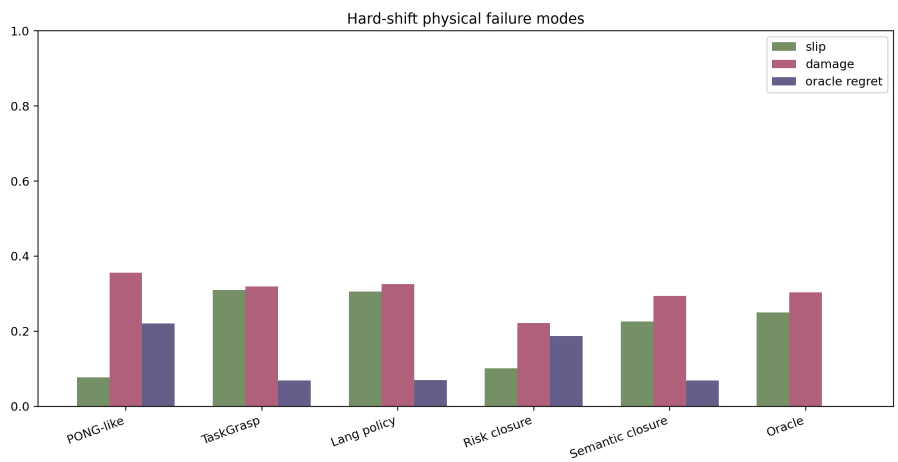

Figure 5: Physical failure modes on combined hard shift. Risk-aware force closure has lower damage, while learned semantic baselines have lower semantic violation.